DR. D. Y. PATIL INSTITUTE OF ENGINEERING, MANAGEMENT AND RESEARCH Akurdi, Pune-411044 Department-Artificial Intelligence and Data Science		
Laboratory Journal | Subject: Computer Laboratory – III		**AY:2025-26**
Name: Atharva Velhankar	**Class:** B.E. (A)	**Roll No:** 22508

Lab Practical 1

Design a distributed application using RPC for remote computation where client submits an integer value to the server and server calculates factorial and returns the result to the client.

Code:

server.py

```
from xmlrpc.server import SimpleXMLRPCServer
from xmlrpc.server import SimpleXMLRPCRequestHandler

class RequestHandler(SimpleXMLRPCRequestHandler):
    rpc_paths = ('/api/v1/rpc_handler',)

class RPCServer:
    def calculate_factorial(self, n):
        if n < 0:
            raise ValueError("Input must be a non-negative integer.")
        result = 1
        for i in range(1, n + 1):
            result *= i
        return str(result)  # convert to string before sending

with SimpleXMLRPCServer(('localhost', 8000), requestHandler=RequestHandler) as server:
    server.register_introspection_functions()
    server.register_instance(RPCServer())
    print("Server is running on port 8000...")
    server.serve_forever()
```

client.py

```
import xmlrpc.client

with xmlrpc.client.ServerProxy("http://localhost:8000/api/v1/rpc_handler") as proxy:
    try:
        n = int(input("Enter a number: "))
        result = proxy.calculate_factorial(n)
        print(f"Factorial of {n} is: {result}")
    except Exception as e:
        print(f"Error: {e}")
```

Output:

Server:

```
C:\Users\athar\OneDrive\Desktop\Practical\Computational Intelligence\CL-III Pract\pract1>python server.py
Server is running on port 8000...
127.0.0.1 - - [18/Mar/2026 16:14:29] "POST /api/v1/rpc_handler HTTP/1.1" 200 -
```

Client:

```
C:\Users\athar\OneDrive\Desktop\Practical\Computational Intelligence\CL-III Pract\pract1>python client.py
Enter a number: 9
Factorial of 9 is: 362880
```

Sign: _____.

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Lab Practical 2

Distributed String Concatenation Application using RMI.

Code:

StringRemote.java

```
import java.rmi.Remote;
import java.rmi.RemoteException;

public interface StringRemote extends Remote {
    String concatenate(String s1, String s2) throws RemoteException;
}
```

StringRemoteImp.java

```
import java.rmi.RemoteException;
import java.rmi.server.UnicastRemoteObject;

public class StringRemoteImpl extends UnicastRemoteObject implements StringRemote {

    StringRemoteImpl() throws RemoteException {
    }

    public String concatenate(String s1, String s2) {
        return s1 + s2;
    }
}
```

Server.java

```
import java.rmi.registry.LocateRegistry;
import java.rmi.registry.Registry;

public class Server {
    public static void main(String[] args) {
        try {
            StringRemoteImpl obj = new StringRemoteImpl();
            Registry registry = LocateRegistry.createRegistry(6000);
            registry.rebind("StringService", obj);
            System.out.println("Server is running and ready...");
        } catch (Exception e) {
            System.out.println("Server Error: " + e.getMessage());
        }
    }
}
```

Client.java

```
import java.rmi.registry.LocateRegistry;
import java.rmi.registry.Registry;
import java.util.Scanner;

public class Client {
    public static void main(String[] args) {
        try {
            Scanner sc = new Scanner(System.in);
            System.out.print("Enter String 1: ");
            String s1 = sc.nextLine();
            System.out.print("Enter String 2: ");
            String s2 = sc.nextLine();
            Registry registry = LocateRegistry.getRegistry("localhost", 5000);
            StringRemote stub = (StringRemote) registry.lookup("StringService");
            String result = stub.concatenate(s1, s2);
            System.out.println("Concatenated Result: " + result);
            sc.close();
        } catch (Exception e) {
            System.out.println("Client Error: " + e.getMessage());
        }
    }
}
```

Output:

Server:

```
Console ×
Server [Java Application] C:\Users\athar\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_17.0.8.v20230831-
Server is running and ready...
```

Client:

```
Console ×
<terminated> Client [Java Application] C:\Users\athar\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.w
Enter String 1: Hello
Enter String 2: DYPIEMR
Concatenated Result: Hello DYPIEMR
```

Sign: _____.

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Lab Practical 3

Distributed Application using MapReduce for Text Analysis:

input.txt

```
pract3 > input.txt
English is simple.
English is fun.
```

a. Character counting in a given text file.
Code:

char_count.py

```
from collections import Counter
with open("input.txt", "r") as f:
    text = f.read().lower()

# Mapper phase
print("MAPPER OUTPUT")
print("=" * 30)
mapped = []
for char in text:
    if char.strip() != "":
        print(f"{char}\t1")
        mapped.append(char)

# Reducer phase
print("\nREDUCER OUTPUT (Final Count)")
print("=" * 30)
char_count = Counter(mapped)
for char, count in sorted(char_count.items()):
    print(f"{char}\t{count}")
```

Output:

```
C:\Users\athar\OneDrive\Desktop\Practical\Computational Intelligence\CL-III Pract\pract3>python char_count.py
MAPPER OUTPUT
==============================
e	1
n	1
g	1
l	1
i	1
s	1
h	1
i	1
s	1
s	1
i	1
m	1
p	1
l	1
e	1
.	1
e	1
n	1
g	1
l	1
i	1
s	1
h	1
i	1
s	1
f	1
u	1
n	1
.	1

REDUCER OUTPUT (Final Count)
==============================
.	2
e	3
f	1
g	2
h	2
i	5
l	3
m	1
n	3
p	1
s	5
u	1
```

b. Counting no. of occurrences of every word in a given text file.
Code:

<u>word_count.py</u>

```
from collections import Counter

with open("input.txt", "r") as f:
    text = f.read().lower()

# Mapper phase
print("MAPPER OUTPUT")
print("=" * 30)
mapped = []
for word in text.split():
    print(f"{word}\t1")
    mapped.append(word)

# Reducer phase
print("\nREDUCER OUTPUT (Final Count)")
print("=" * 30)
word_count = Counter(mapped)
for word, count in sorted(word_count.items()):
    print(f"{word}\t{count}")
```

Output:

```
C:\Users\athar\OneDrive\Desktop\Practical\Computational Intelligence\CL-III Pract\pract3>python word_count.py
MAPPER OUTPUT
============================
english 1
is      1
simple. 1
english 1
is      1
fun.    1

REDUCER OUTPUT (Final Count)
============================
english 2
fun.    1
is      2
simple. 1
```

Sign: _____.